$$m = [-1, 2]$$

$$b = [1, 1]$$

$$x = ((1,3), (4,10))$$

$$y = (5, 6)$$

x_1	x_2	y	$\hat{y}$
1	3	5	6
4	10	6	17

$$\hat{y} = m_1 x_1 + m_2 x_2 + b = xm + b$$

$$\hat{y} = xm + b$$

$$\begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 + 6 \\ -4 + 20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$mse = E(\hat{y} - y)^2$$

$$\begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}^2 = \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \frac{1 + 121}{2}$$

$$= \frac{122}{2}$$

$$= 61$$

$$let\ u = xm + b - y$$

$$m = \frac{2 \times Tn\ (\hat{y} - y)}{n}$$

$$E = \frac{u^2}{n}$$

$$\frac{dE}{du} = \frac{2u}{n}$$

$$\frac{du}{dm} = x^T$$

$$\frac{dE}{dm} = \frac{2ux^T}{n} \rightarrow \frac{2x^T(\hat{y} - y)}{n}$$

$$= x^T \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix}^T = \begin{bmatrix} 1 + 44 \\ 3 + 110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \frac{1}{100} \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} -1 - \frac{45}{100} \\ 2 - \frac{113}{100} \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$E = \frac{\Sigma (\hat{y} - y)^2}{n} = \frac{u^2}{n}$$

$$u = \hat{y} - y$$

$$\frac{dE}{du} = \frac{2u}{n}$$

$$\frac{du}{db} = (\hat{y} - y)' = (\cancel{mx} + b - y)$$

$$= 1 \quad \frac{dE}{db} = \frac{2y}{n}$$

$$= \frac{2(\hat{y} - y)}{n}$$

$$\frac{dz}{db} = \cancel{2}\left[\begin{matrix}6\\17\end{matrix}\right] - \left[\begin{matrix}5\\6\end{matrix}\right]$$

$$= \left[\begin{matrix}1\\11\end{matrix}\right]^{2\cancel{x}}$$

$$= 12$$

$$b_{new} = \left[\begin{matrix}1\\1\end{matrix}\right] - \left[\begin{matrix}1\\100\end{matrix}\right]\left[\begin{matrix}12\\12\end{matrix}\right]$$

$$= \left[\begin{matrix}1\\1\end{matrix}\right] - \left[\begin{matrix}0.12\\0.12\end{matrix}\right]$$

$$= \left[\begin{matrix}0.88\\0.88\end{matrix}\right]$$